

Summary

- Migratory Stopover Planning Tool derived from Audubon data. [In a shiny app here](#) - has stopover data for CERW, GWWA, WOTH

April 14

Road to Recovery History

Held at 111 Massachussetts Avenue - renovated building, now home to Georgetown new Institute.

```
nodes:
  organization-road-to-recovery:
    short_key: r2r
    title: Road to Recovery
    type: organization
    description: Non-profit formed to address issues highlighted in the 3 Billion
    Birds paper.
    tags: [non-profit, birds, 3 billion birds]
```

Workshop Goals

Katie Holland

Goals

1. Explore the R2R process for Eastern Tipping Point Species (and others)
2. Explore the topic of migratory stopover habitat conservation

R2R Process

- limiting factors during migration?
 - where on the journey?
 - what is the threat?
 - When is the impact? When is the population impacted?
 - How significant is the impact?

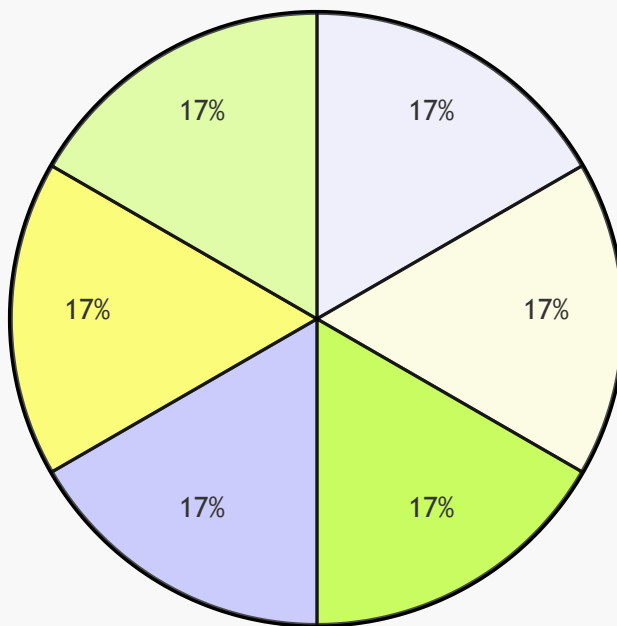
Day 1 Agenda

- R2R Framework and Recovery Planning
- Spp Working Group Breakouts Lunch
- Migratory Stopover Tools
- Discussions

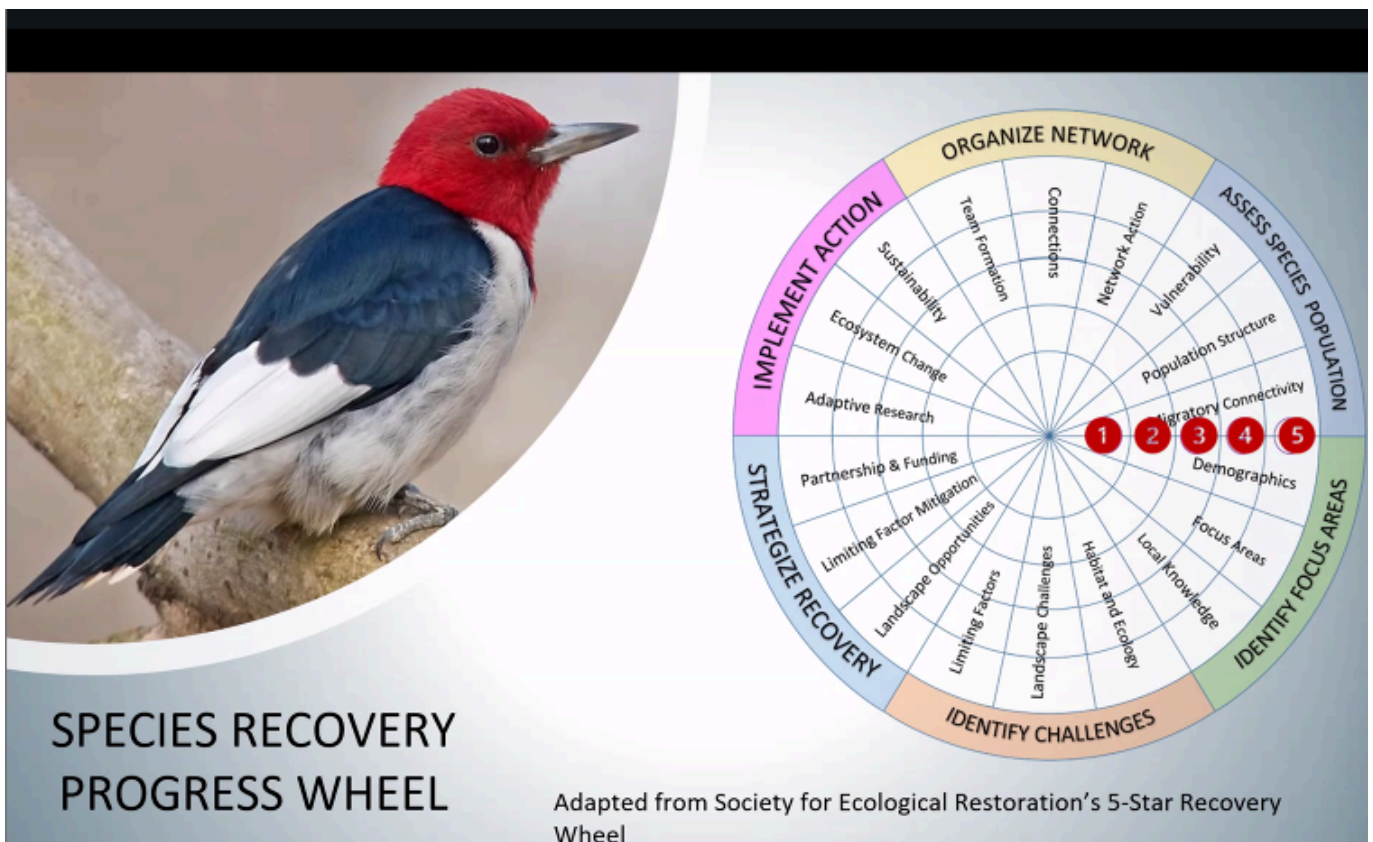
R2R Recovery Progress Wheel

Intended to track recovery of species, not evaluate or compare Working Groups. For tipping point species, not recovered, so wheel by definition will be incomplete.

R2R Recovery Wheel

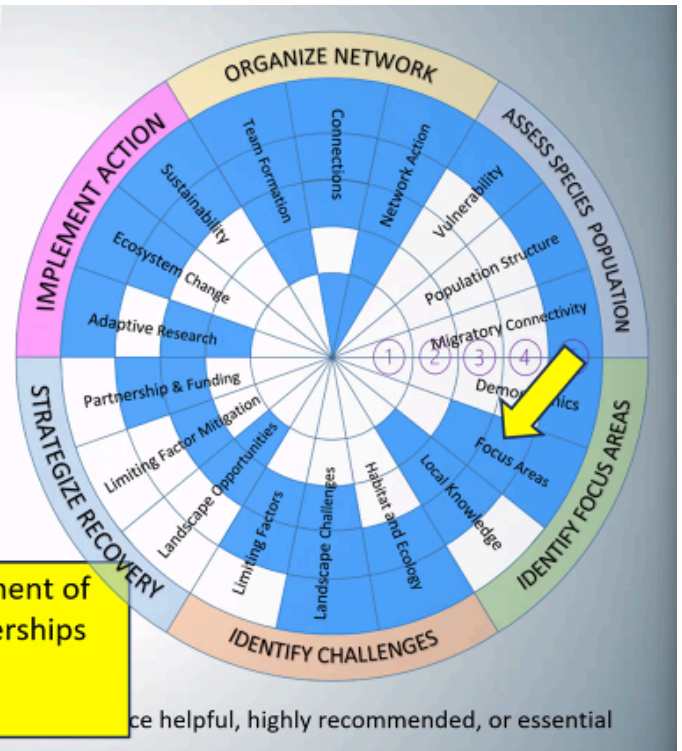


- Assess Species Population [16.67]
- Identify Focal Areas [16.67]
- Identify Challenges [16.67]
- Strategize Recovery [16.67]
- Implement Action [16.67]
- Organize Network [16.67]





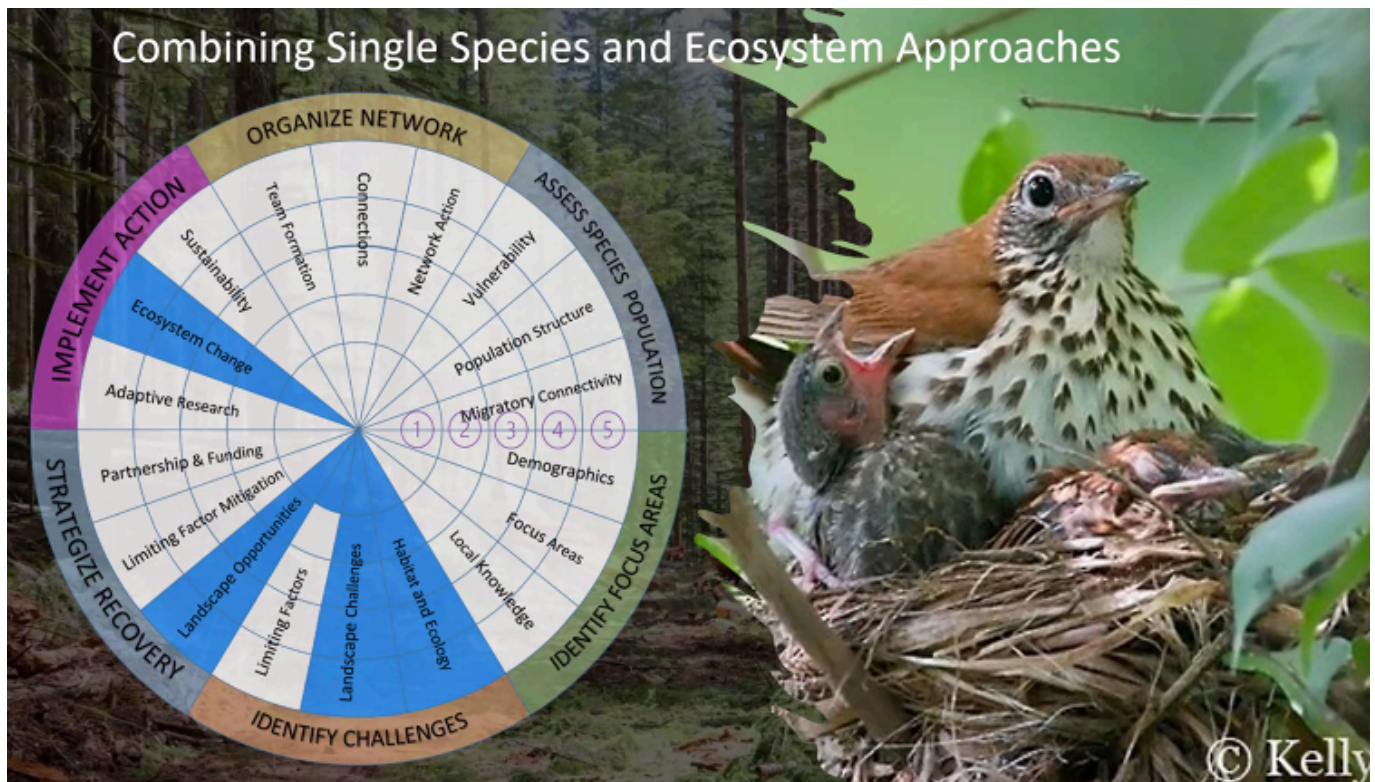
for EXAMPLE— 3.2.4. After biological assessment of focus areas, network analysis suggests partnerships that might be most promising for leveraging conservation action (not just bird-oriented).



If you don't have a network, can't do a network analysis!

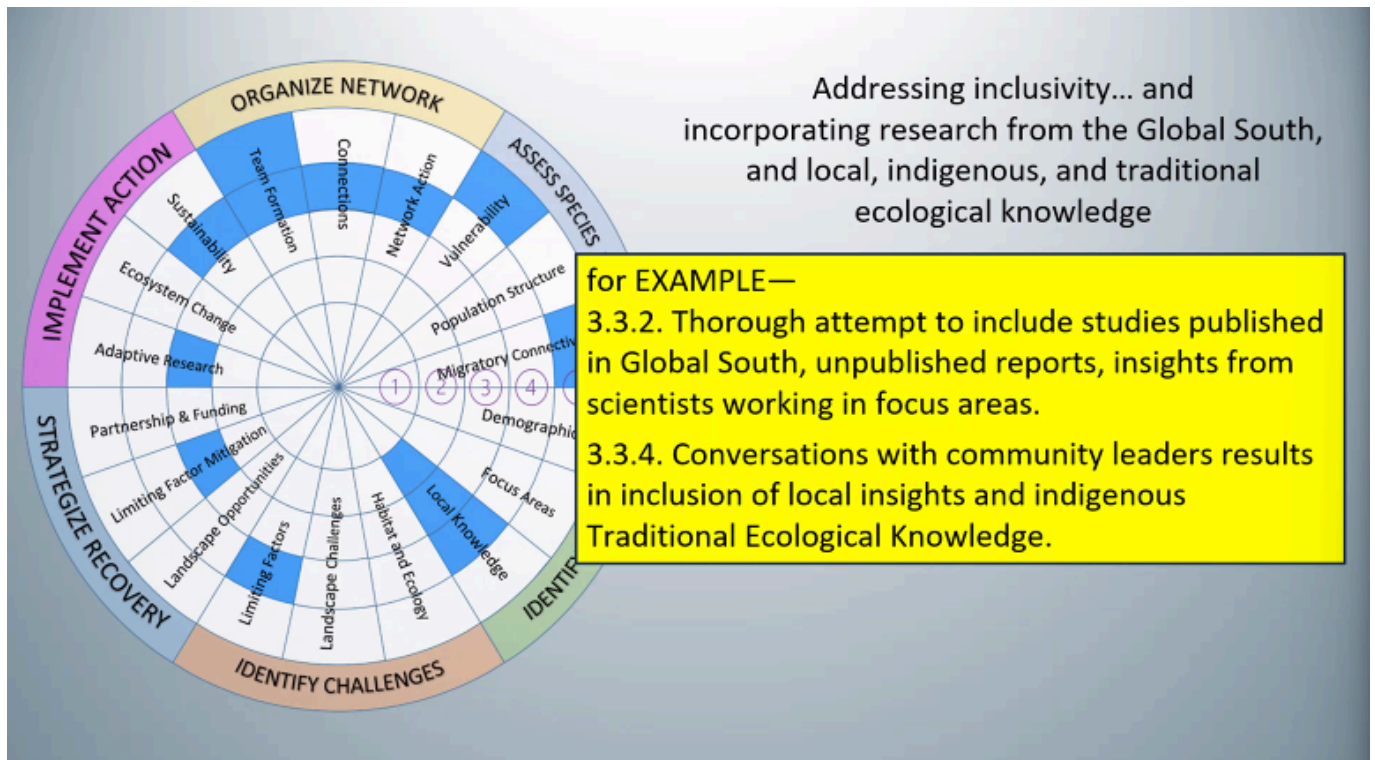
Diagrams are designed to go from inside (1) to outside (5), but this does not always make sense in term of development.

Combining Single Species and Ecosystem Approaches

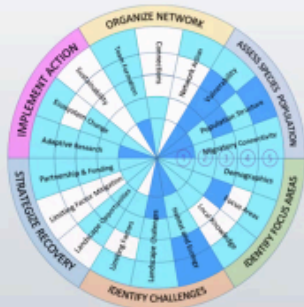


Focus on 1-3 this morning.

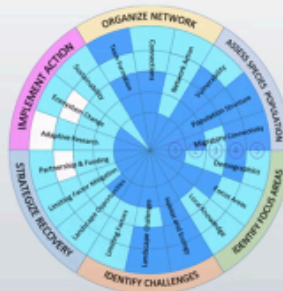
Addressing Inclusivity



SHARED PROGRESS AND NEED



Cerulean Warbler

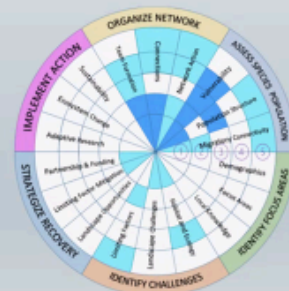


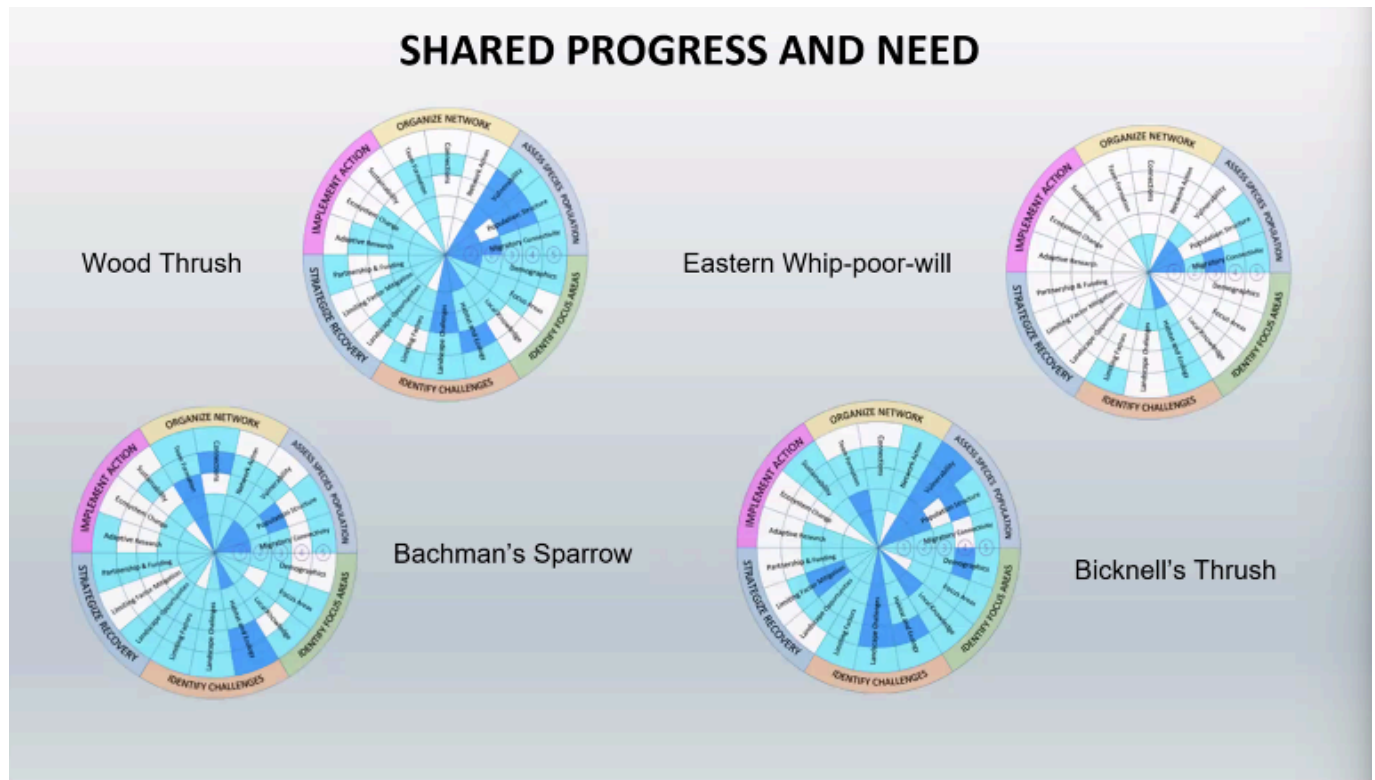
Golden Winged Warbler

Evening Grosbeak



Connecticut Warbler





Assignment for today

Not updating the wheel - should be done at another time.

Today, look at completed wheels, and prioritize what is next for the group.

Note: organize network is a common problem - 1.1.5 Group organizational plan and clear rules of engagement established.

Mural Board

EWPW Working Group breakout

See [mural board](#) with information.

- Initial focus on building out network?
 - Develop list of names of people who might be interested? Be mindful of the ask for help/resources.
- Cultural significance of this species in the Ozarks - elsewhere?
- Peele: Nexus with Hurricane Helene recovery efforts?
- Not a lot of knowledge among group with the [Nightjar Survey Network](#)
- Leader is [Christopher M Tonra](#)

Migratory Stopover Tools

Randy Dettmers (USFWS)

Overview

- latest tech/monitoring efforts
- gather feedback about questions about state of knowledge/comfort

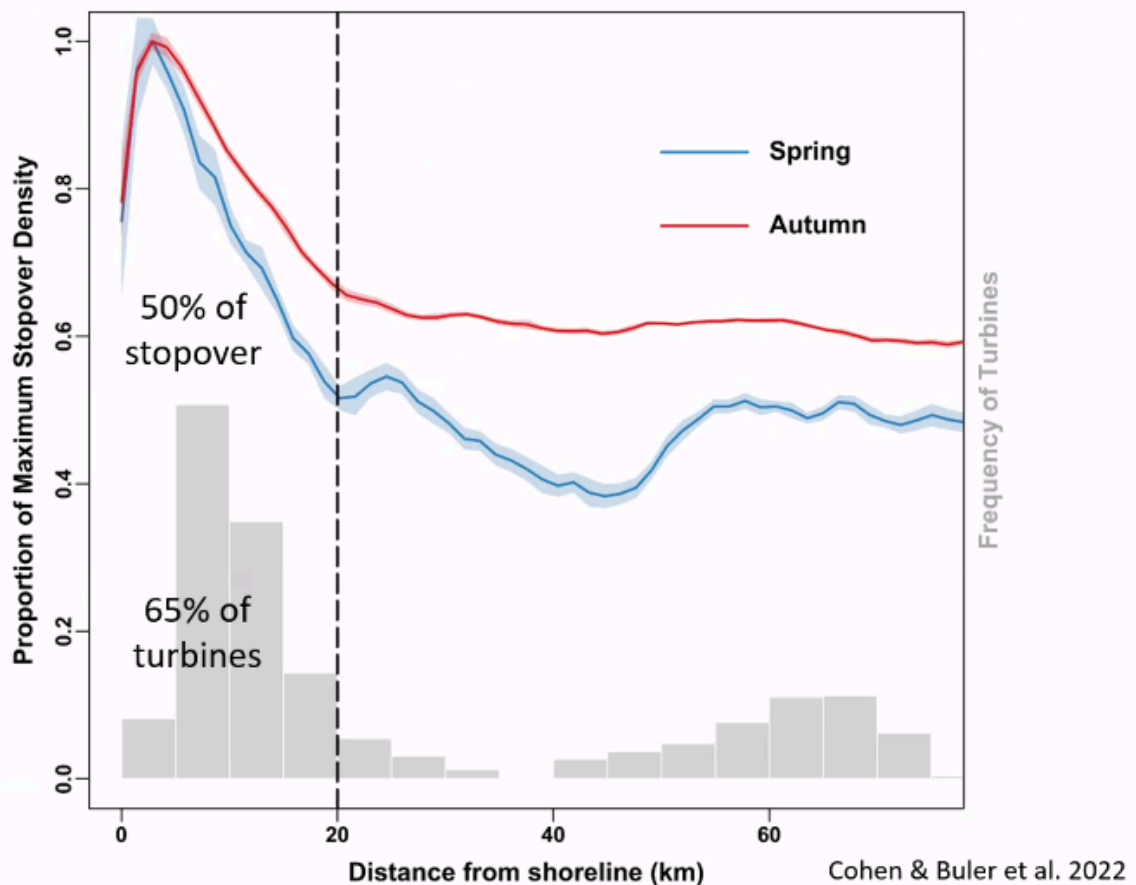
- do we know enough to start doing something?

Aeroecology Program at U of Delaware

Jeffrey Buller

- 1/3 of US covered by radar - continuous data for >30 years
- Since 2008, 250m pixel resolution
 - scan every 5-10 minutes
- During migration, probably more biology than meteorology in the data.
- [Stopover Hotspot Patterns and Protection Status](#)

Stopover migrants concentrate near coastlines



- Some radar networks in Mexico, Brazil, not as openly available.
- Canada working to make their network available.

Mapping stopover hotspots for state-level conservation planning

Claire Nemes cenemes@gmail.com claire.nemes@umce.edu

Mapping stopover hotspots for state-level conservation planning

Claire E. Nemes, Gwenda L. Brewer, Fengyi Guo, Jeffrey J. Buler, Sergio A. Cabrera-Cruz, Brian R. Tsuru, Emily B. Cohen

Road to Recovery & Partners in Flight Eastern Working Group Regional Workshop

2026-04-14
Georgetown University
Washington, DC



University of Maryland
CENTER FOR ENVIRONMENTAL SCIENCE
APPALACHIAN LABORATORY



- Lots of great research with advancement of technology
- Comes with a sizeable research-implementation gap
 - [Fabian et al 2019](#), [Kadykalo et al 2021](#)
- Looked at SWAPs

How biologists use information in State Wildlife Action Plans (SWAPs)

Habitat management

Restoration, improvement, threat reduction on public & private lands

Land acquisition & easements

Target new locations for conservation

Legal testimonies

Testifying in favor of / opposition to (e.g.) proposed development

Funding proposals

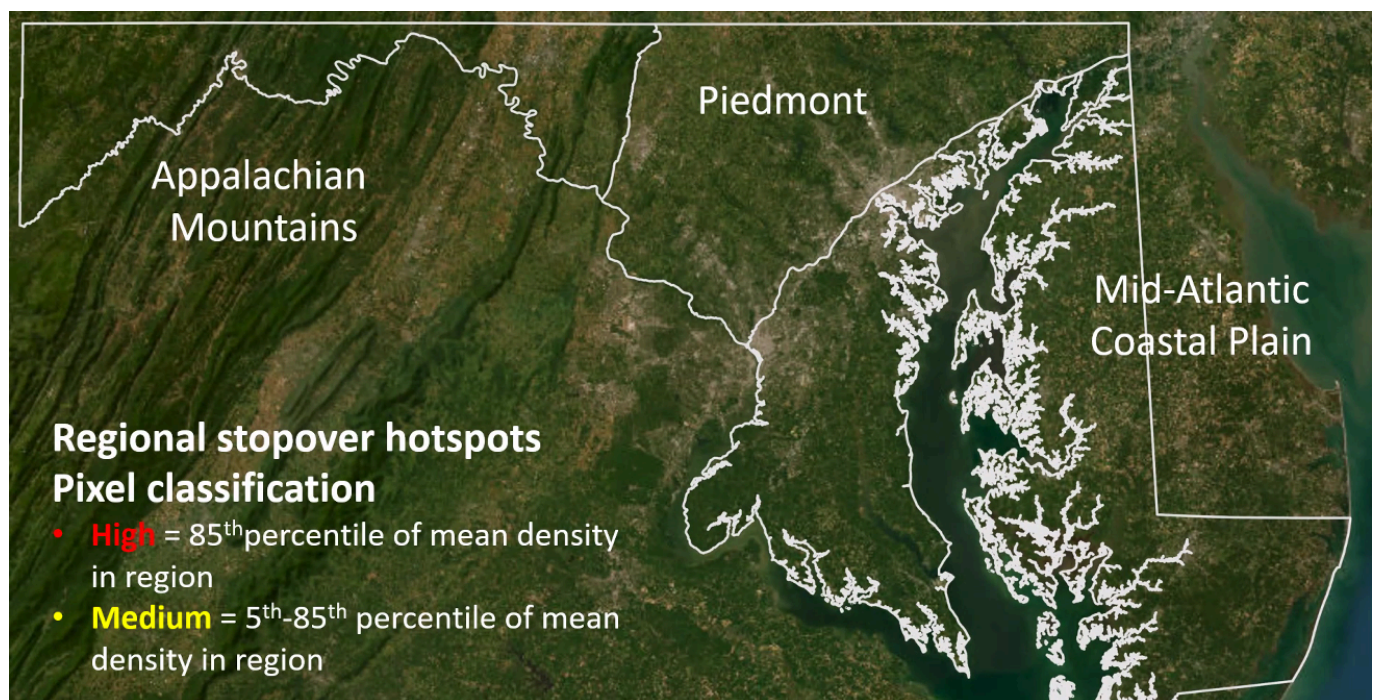
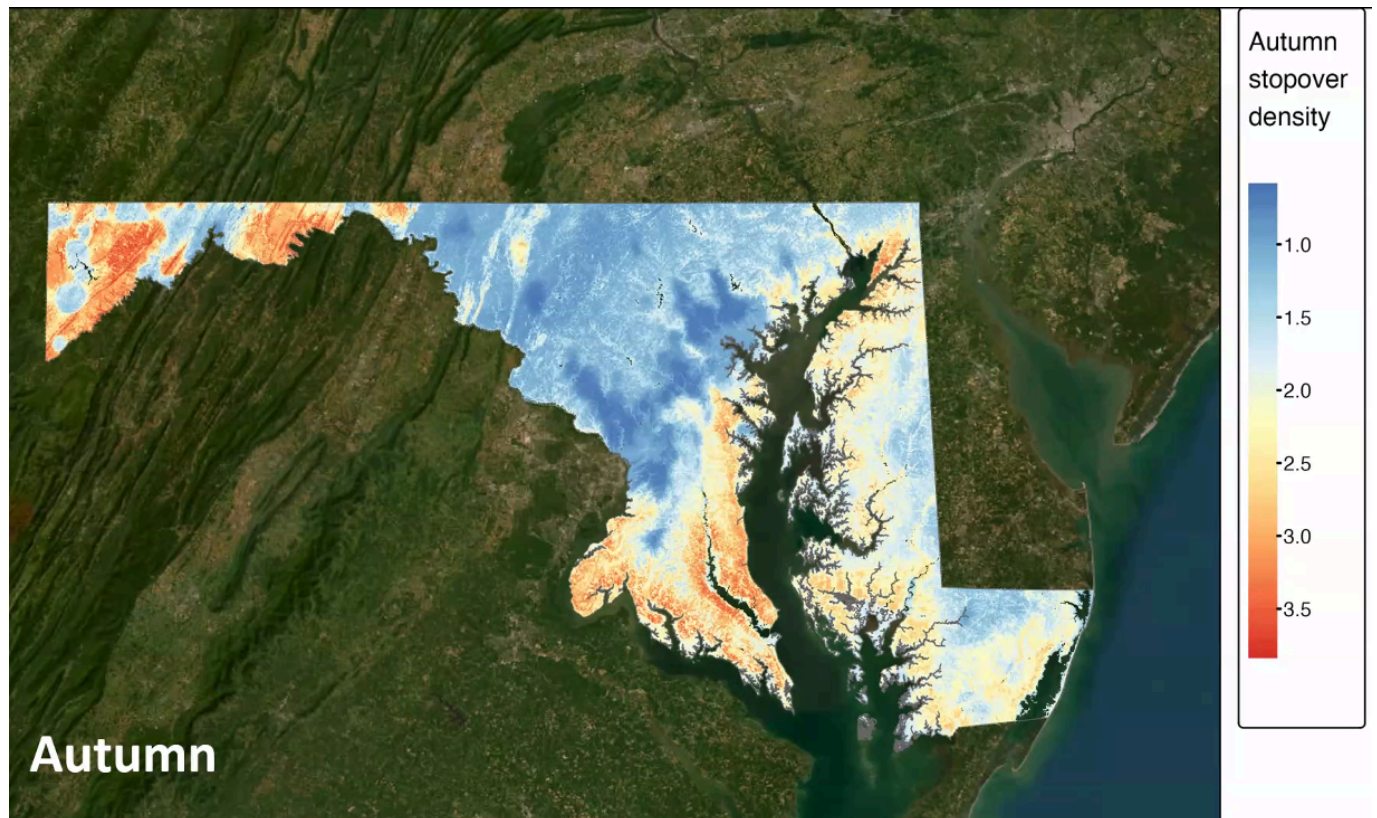
Supporting evidence in grant proposals

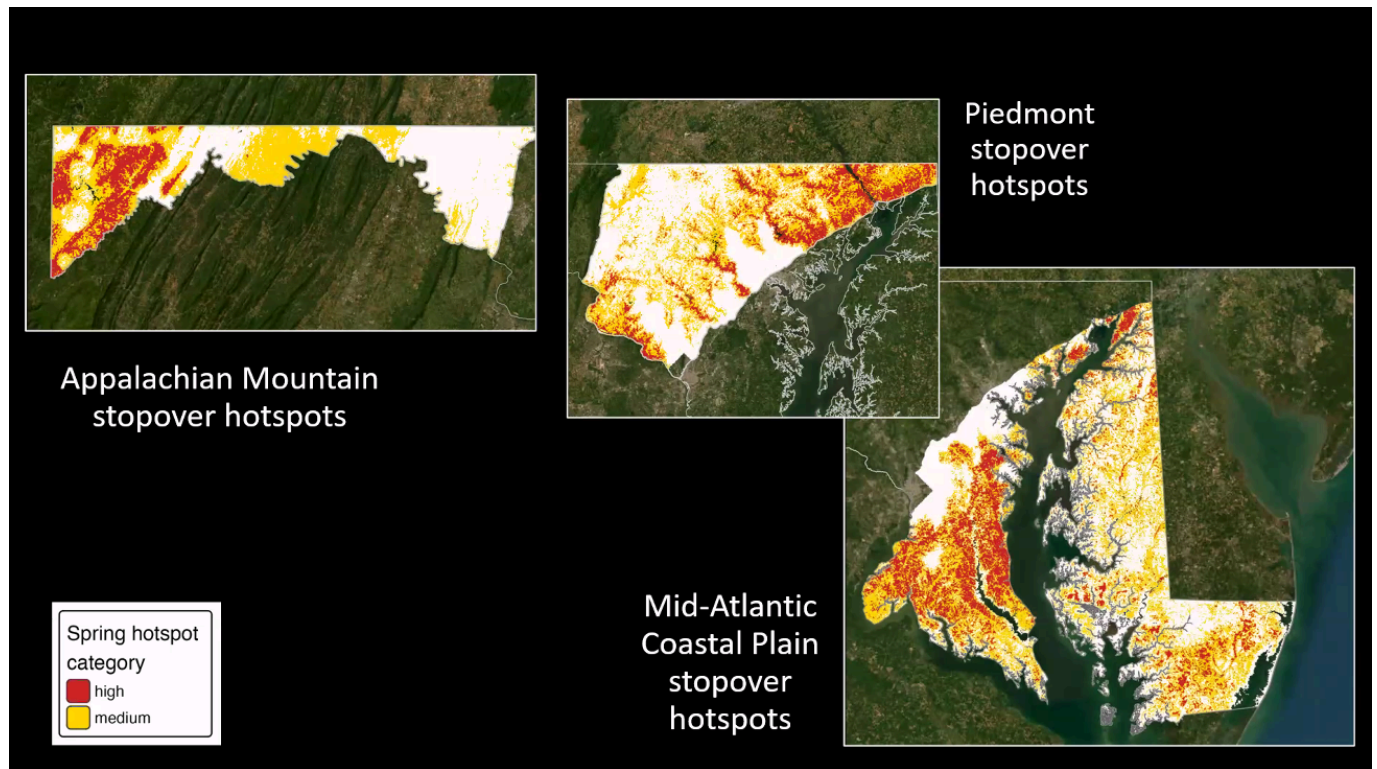
Outreach

Community engagement and connection to place

- most SWAP focus on breeding season, but MD has also worked for migrants.

Example Data





Using Multi-phase migration Models to Inform Conservation Planning

Bill Deluca, Director of Migratory Science [Ryan Brodie](#), University of Maine [Paper](#)

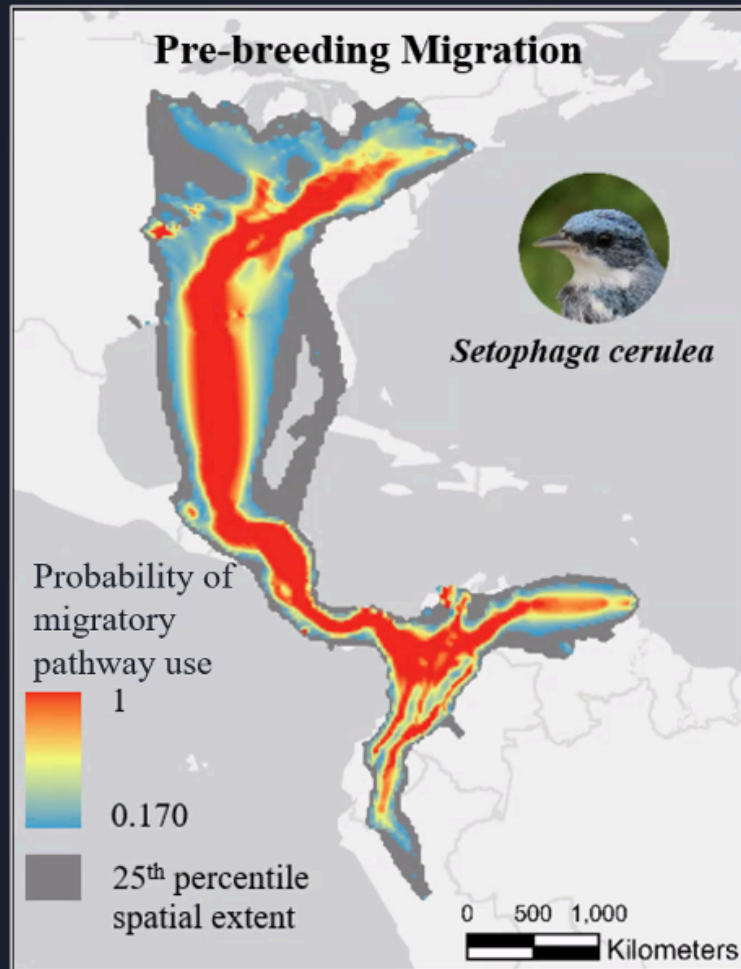
Merged Migration Maps

- integrated full annual cycle data from tagging, eBird, band data
- working to incorporate Motus
- working to integrate active stopover sites (not considered in v1.0)

Species examples

- GWWA, CERW, WOTH, CAWA

Phase I: Migratory Pathways



Probability of pathway use for Cerulean warblers during pre-breeding migration

Feel free to reach out:
ryan.brodie@maine.edu

Link to paper:
<https://doi.org/10.1093/ornithapp/duag030>



Bird Concentration Areas: A Framework to identify important locations for bird communities during migration

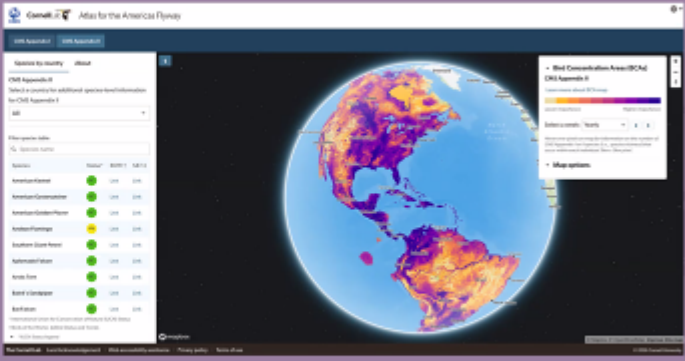
[Fabiola Rodriguez-Vasquez](#)

- to identify migratory bird communities, we need distribution data across space and time
- Essentially stacking eBird Status Layers to identify highest value of bird presence, sum to bird concentration indices.
- Identify areas that support a high number of individuals
- Identify areas that support a high number of species
- Delineating BCAs for forest-associated, migratory species

2

Delineating BCAs for the UN- CMS





Ten species that are a focus of this workshop:

Cerulean Warbler

Eastern Towhee

Evening Grosbeak

Golden-winged Warbler

Prairie Warbler

Wood Thrush

Connecticut Warbler

Bicknell's Thrush

Chimney Swift

Bachman's Sparrow

Eastern Whip-poor-Will

BCAs are a framework to identify important locations for bird communities during migration.

Advantages

Spatial-temporal flexibility

Single-species/Communities

Updates ~ 2 years, add species

Challenges

With limits for stopover <1 week

Data availability/quality varies by spp

Does not provide information on habitat quality based on individual's performance

Break

Muyral



NCASI Project

Lots of effort put into ground truthing data, including vegetation and formal bird surveys.

High expense for commercial forests.

April 15

Demonstrations: Bird Conservation Areas

Fabiola Rodriguez (Fabi)

[See Mural for links/examples](#)

- mapping Bird Concentration Index richness and abundance for migration...

Migratory Bird - Conservation Planning Tool

[Ryan Brodie](#) - U of ME

- [R Shiny Tool](#)
- [Also available on the Upper Miss JV](#)
- Migratory Pathways - Bill Deluca and Ryan
 - simplified version of Audubon movement tracking, least-cost paths, etc.
- Some interpolation about pre- and post- breeding timing.

Seasonal Exposure to hemispheric Conservation Challenges Influences Population Trends of Migratory Warblers

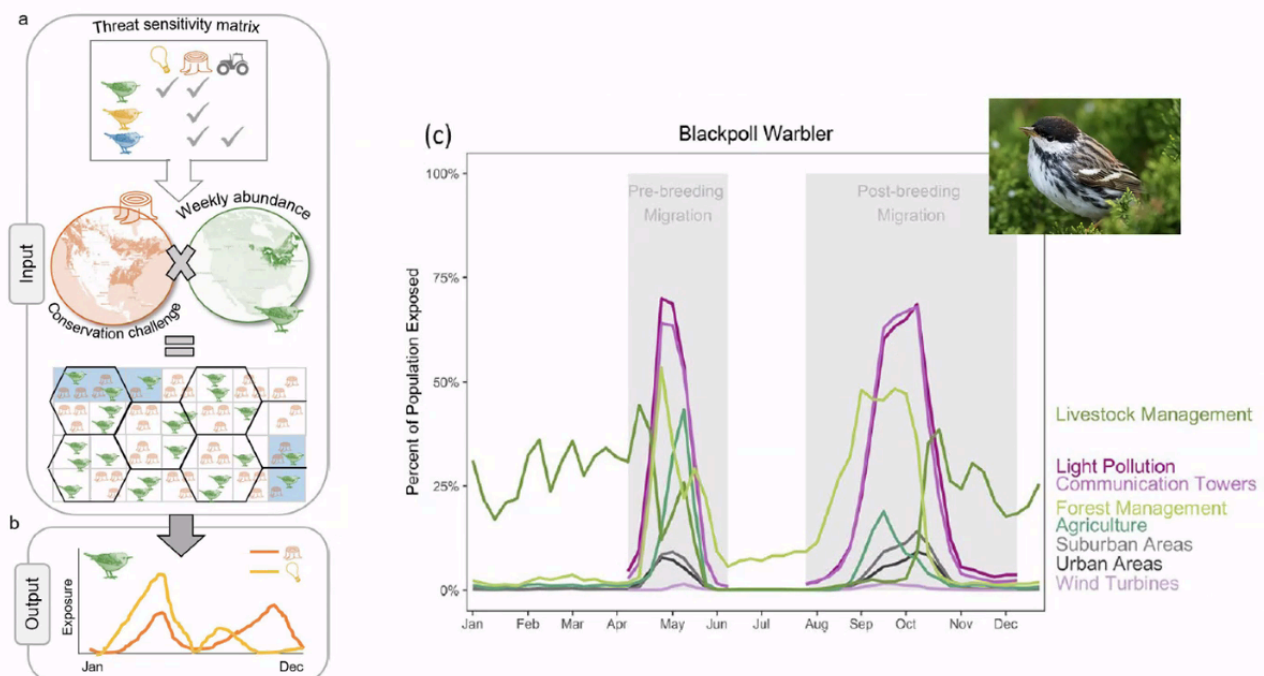
[Jeffrey Ritterson - National Audubon]

- Audubon Bird Migration Explorer
- 19 challenges, 9 categories
- 458 species, 88 standardized threats, Stresses (7 from IUCN)
 - [Threat Sensitivity Matrix](#) - not shareable yet.
- In general, post-breeding migration was biggest mortality risk for spp.
-

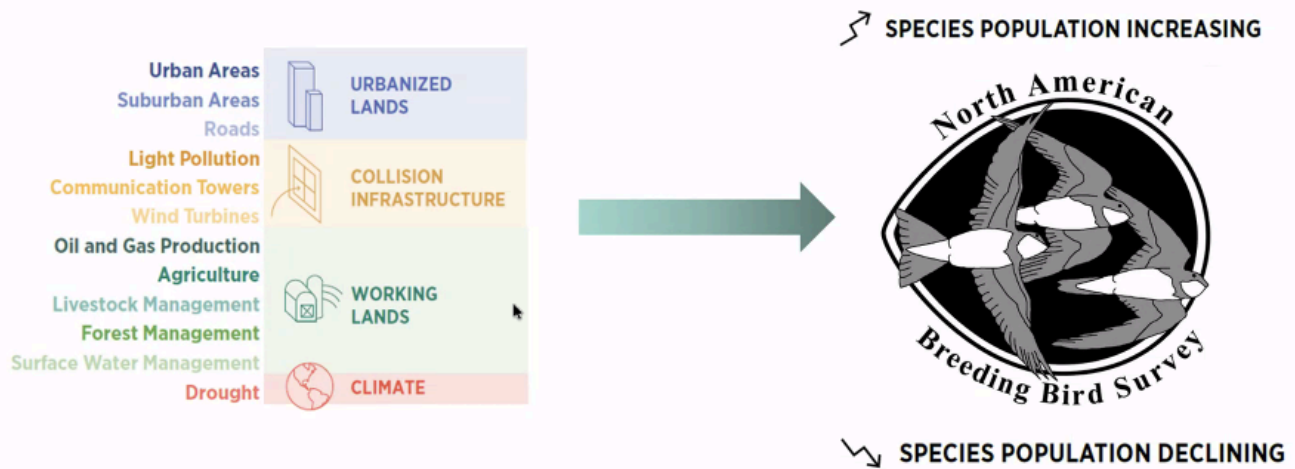
- 1 Literature review to link species to conservation challenges.
- 2 Estimate warbler exposure to conservation challenges across the annual cycle.
- 3 Determine the extent to which threat exposure drives population trends.

Seavy and Smith et al. 2025. Global Change Biology
<https://doi.org/10.1111/gcb.70121>

Exposure across the full annual cycle



How important is exposure to conservation challenges when explaining warbler population trends?



Human Dimensions

[Katie Holland]

- Focus Area Profile - develop broad context for needs
 - Embedded in wheel
- Refer to Action category 3 "Identify Focus Areas"
 - largely HD focused
 - Integration is critical here.

Key Components of the Wheel



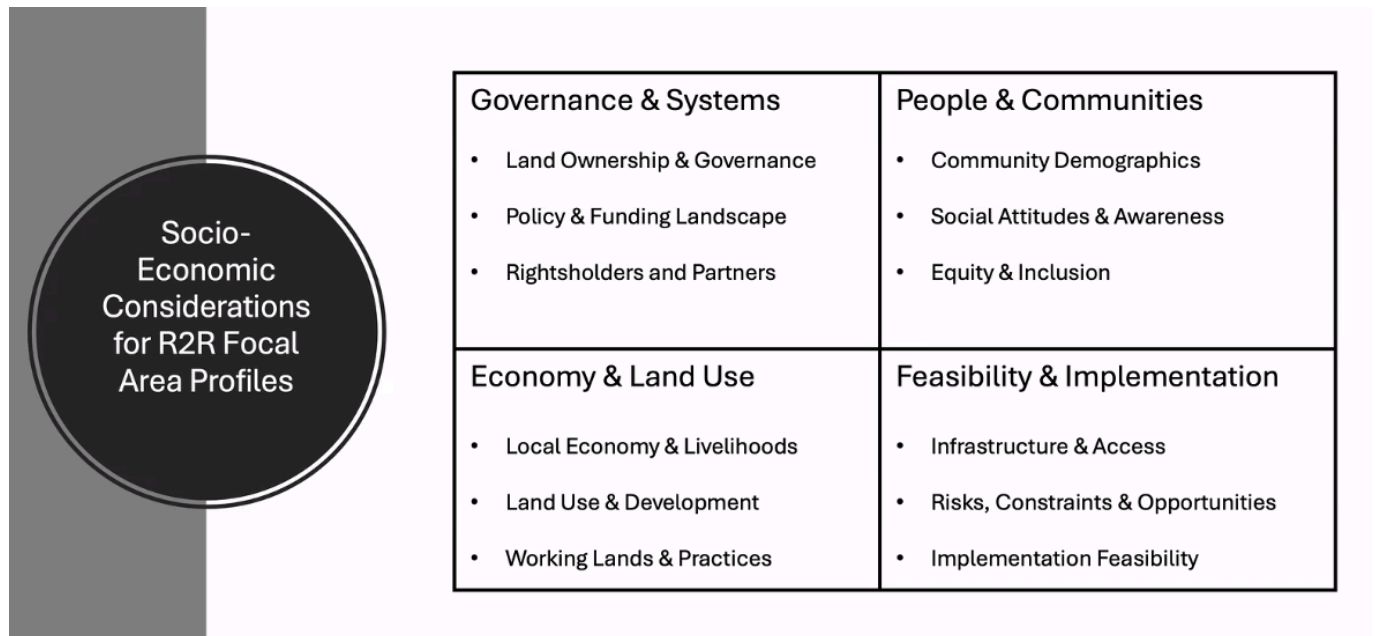
Within Identify Focus Areas...

3.1.5 Socioeconomic correlates of demographic parameters identified

3.2.5 Biological focus areas refined by including human landscape assessment

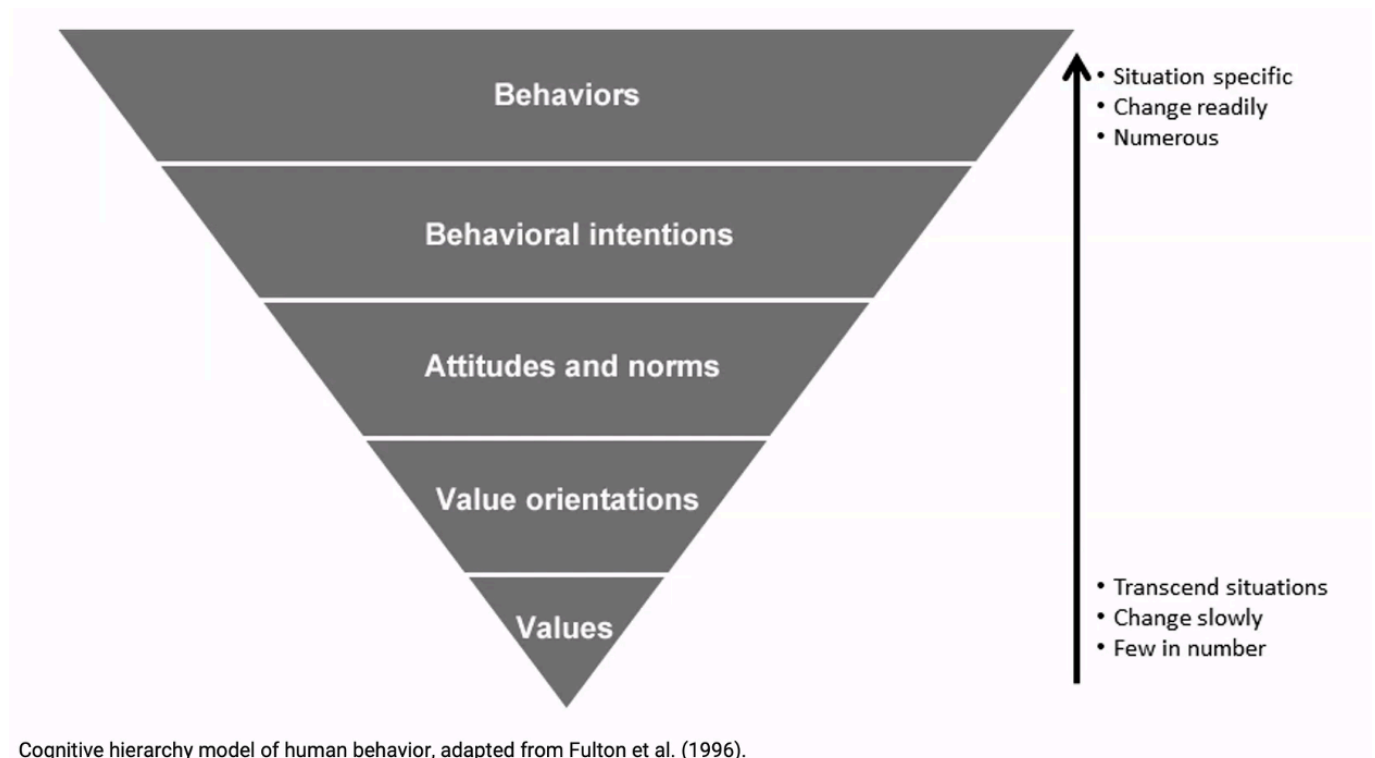
3.3.1 - 3.3.4 Local Knowledge

3.3.5 Focus Area Profile Complete



Social Attitudes and Awareness

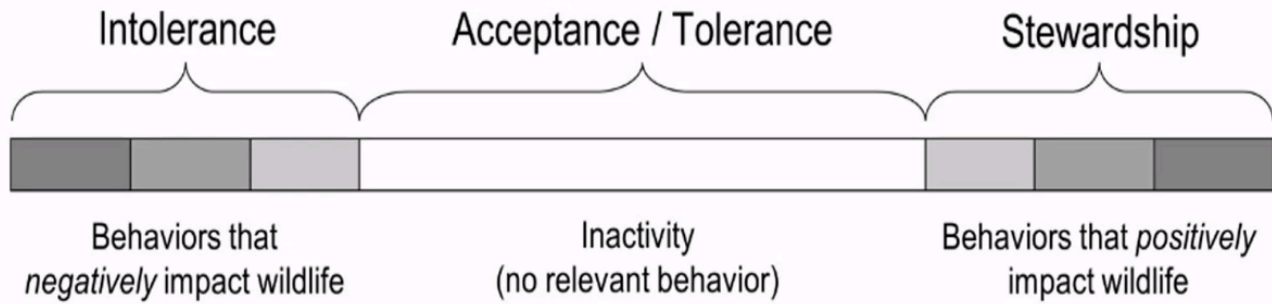
- "How do we get people to care?"
- Birding numbers off the charts, how do we get them to care (usually meaning behaviors)?



- it is really hard to change values - easier to change behaviors
- Outreach doesn't change values

Wildlife Value Orientation Changes

- Shifting away from utilitarian towards mutualism



Bruskotter and Fulton, 2012; Bruskotter and Wilson, 2014; Bruskotter et al., 2015)

- typically, goal is tolerance, but for tipping point species, we need more.
-